Muhammad Aitessam Ahmed

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Senior AI Engineer | LLMs, RAG and Agentic AI | Healthcare AI

Nationality: Pakistani | Open to relocation (Gulf, Europe, UK) or remote

PROFESSIONAL SUMMARY

Senior AI Engineer with 5+ years taking AI into production in healthcare and research: RAG with hybrid search and cross-encoder re-ranking, LLM document extraction that cut manual data entry by 60-70%, multi-agent workflows (LangGraph, CrewAI, MCP), clinical NLP and speech-to-text, and, most recently, a research-stage patient digital twin and PBPK drug simulation engine. Python first, deploying with FastAPI, Docker and Kubernetes on AWS, Azure and GCP.

TECHNICAL SKILLS

Core AI/ML: LLMs, RAG (BM25 + vector + cross-encoder reranking), Agentic AI, Generative AI, NLP, Computer Vision, Deep Learning, Fine-Tuning, Multimodal AI, Embeddings, LSTM

MLOps & Infra: FastAPI, Docker, Kubernetes, AWS, Azure, CI/CD, GPU Inference, REST APIs, Async Python, Model Serving, Compliance-Grade Deployment, Celery, GCP, pytest

Frameworks: LangGraph, MCP, PyTorch, TensorFlow, LangChain, AGNO, CrewAI, HuggingFace, scikit-learn, OpenCV

Data & Storage: FAISS, Pinecone, Chroma, Weaviate, PostgreSQL, MongoDB, MySQL, Elasticsearch

Scientific Modelling: Digital Twins, PBPK Modelling, PK/PD (Hill/Emax), ODE Modelling (SciPy solve_ivp), XGBoost, SHAP, MLflow, PubChem (PUG REST), Synthetic Cohort Generation

Tooling: n8n, Power BI, Tableau, Kafka, Google Vision API, Vertex AI

Languages: Python (primary), SQL, C/C++

WORK EXPERIENCE

Senior AI Engineer | BlockMed Pro, Islamabad, PK (remote)

May 2026 - Present

Research-stage patient digital twin and in-silico drug simulation platform for pharma studies in type 2 diabetes (not clinically validated).

- Built the metabolic core and PBPK engine of a multi-organ patient digital twin for type 2 diabetes, coupling published glucose-insulin (Bergman/Dalla Man), kidney (CKD-EPI 2021) and cardiovascular (Heldt 2002) models with liver, gut, adipose and CNS modules through a defined organ-exchange interface in Python (NumPy, SciPy solve_ivp with LSODA/Radau); simulated a four-arm, 282-patient virtual cohort (564 runs) in about 13 minutes, backed by ~790 automated tests.
- Built a safety-gated drug registry and PK/PD sourcing pipeline (PubChem PUG REST properties, Rodgers-Rowland tissue partitioning, transit-compartment absorption, Hill/Emax dose-response with a tiered fallback), validated against published clinical data: corrected metformin Tmax from 0.08 h to ~3 h, inside the label range, and gliclazide passed 8 of 8 validation checks with Cmax within 12% of the published value.
- Built FastAPI and Pydantic services for a pharma study platform (schema mapping across 55 fields and 7 tables, 10-dimension cohort filtering, drug arms, Celery background runs on PostgreSQL), cutting cohort preview latency from ~6 s to under 150 ms with ~330 API tests, and generated a 100,000-patient synthetic cohort in ~81 s calibrated to published UK type 2 diabetes audit figures (NDA, CPRD).
- Surfaced a cross-organ drug interaction in simulation: atenolol lowering heart rate by ~13% reduced cardiac output by ~10%, which slowed renal clearance of metformin and raised its plasma levels.

Senior AI/ML Engineer | Askari Technologies, Islamabad, PK

Aug 2025 - Apr 2026

Built the full AI layer of a live enterprise healthcare product from document ingestion to real-time clinical Q&A serving physicians across multiple specialties.

- Deployed a HIPAA-compliant AI medical assistant to live clinical users, combining RAG-based retrieval and LLM generation with PHI isolation, full audit logging, and output guardrails that prevent sensitive data from entering model context.

- Reduced manual data entry by 60-70% by architecting a Generative AI document processing pipeline that ingests unstructured medical records, runs OCR and LLM-based entity extraction, and outputs validated FHIR R4-compliant resources integrated directly with the hospital EHR.
- Cut physician documentation time by ~50% by shipping a production speech-to-text and clinical NLP pipeline where clinicians dictate and the system generates structured SOAP notes, visit summaries, and prescriptions each reviewed and approved in under 30 seconds.

Machine Learning Engineer | Tensor Labs, Islamabad, PK

Nov 2024 - Jul 2025

Production AI across radiology, large-scale image intelligence, fashion tech, and enterprise workflow automation, multiple live client products serving real users.

- Reduced radiology review effort by ~40% by designing and deploying a lung nodule classification system with a CNN inference pipeline, LLM-generated clinical reports, and a GPU-served FastAPI service integrated into a live radiology workflow.
- Reduced listing generation and content enrichment turnaround from hours to minutes for real estate and e-commerce clients by deploying multimodal RAG and LLM automation workflows using LangChain and n8n.
- Replaced a fully manual legal compliance workflow by building an image intelligence platform processing thousands of items daily via Google Vision API, async Python pipelines, and structured result storage, removing the need for human review of every item.

AI Research Engineer | Center of Excellence - Artificial Intelligence, Islamabad, PK

May 2022 - Aug 2024

ML engineering for autonomous systems, UAV flight analytics, and predictive reliability in a government-affiliated defence research environment.

- Improved autonomous system reliability by 25% by owning the full lifecycle of an LSTM-based fault detection system, from raw multi-sensor telemetry ingestion and feature engineering through model deployment in an operational environment that flagged failures before they occurred.
- Removed a shared manual preprocessing bottleneck across a team of six engineers by building production ML data pipelines that transformed high-frequency telemetry and simulation data into validated, model-ready datasets, cutting per-experiment setup time from days to hours.

AI Engineer | SKYRONE Technologies, Islamabad, PK

Sep 2021 - Apr 2022

Computer vision and autonomous coordination systems deployed on security and search-and-rescue drone platforms in live field environments.

- Field-deployed a YOLO-based human detection model for autonomous security drones, optimised for on-device edge inference at sub-200ms latency on resource-constrained hardware with no cloud dependency, validated in live outdoor environments across variable conditions.

EDUCATION

BSc Engineering | Institute of Space Technology, Islamabad, PK

2017 - 2021

Thesis: Deep Learning Speech Recognition for Real-Time UAV Control

Designed and deployed an end-to-end speech recognition system for real-time UAV control on low-power edge hardware. Converted raw speech signals to 2D spectrograms and fine-tuned a ResNet50 model (ImageNet pre-trained) on Kaggle GPU infrastructure, achieving 96.45% validation accuracy. Optimised and deployed the inference pipeline on NVIDIA Jetson Nano for on-device real-time command processing with no cloud dependency, successfully controlling quadcopter flight manoeuvres via live speech commands.

PUBLICATIONS & ACHIEVEMENTS

- 1st Place, International Free Mission UAV Competition, Tekno-Fest 2021, Istanbul, Turkey (Team Captain leading a multi-nation field deployment against teams from 30+ countries).
- IEEE Publication: "Design and Development of Audio Processing and Speech Recognition Algorithm," ICASE 2021. DOI: 10.1109/ICASE54940.2021.9904277
- AWS Certified Machine Learning Specialty (In Progress) | Azure AI Engineer Associate (In Progress)